

Cesar Peralta

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Professional Summary

McGill Computer Science (Artificial Intelligence) student with 2+ years building Python automation and clinical data pipelines. Brings healthcare experience to applied AI, data engineering, and backend systems, with hands-on work in LLM agents, RAG, NLP, SQL databases, and Linux/Docker environments, plus undergraduate research experience evaluating an ML benchmark in bioinformatics.

Technical Skills

- **Programming & data:** Python, Java, SQL, C, pandas, Pydantic, ETL, REST/REDCap APIs, CSV, JSON
- **AI/ML & platforms:** RAG, LLM agents, NLP, protein language models, embeddings, ChromaDB, PostgreSQL, LangGraph, MCP/FastMCP, Ollama
- **Development tools:** Docker, Git/GitHub, Linux/Ubuntu, HPC
- **Spoken languages:** English and Spanish (native/bilingual), French (full professional)

Experience

Clinical Data Engineer

Lady Davis Institute for Medical Research | August 2025 – Present

- Automated approximately 90% of a diabetic retinopathy image processing workflow that previously required about 7 hours each week, using Python to control a desktop application, validate conditions, export data, and generate PDF reports.
- Built a rule-based patient report email automation that selects the correct message for each result and clinical scenario, cutting preparation from 2–3 minutes per patient to about one second and reducing manual errors.
- Refactored clinical data pipelines to run faster and be easier to read, maintain, and extend. Added standardized cleaning and privacy safeguards for research datasets.

Undergraduate Researcher

McGill School of Computer Science | May 2026 – July 2026

- Reproduced a published machine-learning benchmark for protein-interaction prediction, establishing a 64.9% test-accuracy baseline before evaluating data-quality and preprocessing changes.
- Ran 13 dataset variants and 90 repeated experiments on a compute cluster; filtering low-confidence training examples raised accuracy from 63.6% to 75.4%.
- Built scalable preprocessing and data leakage checks across two protein-interaction datasets, coordinating parallel experiments with Git branches and worktrees. Results are being prepared for a research manuscript.

Clinical Research Coordinator

Lady Davis Institute for Medical Research | March 2024 – August 2025

- Built Python and REDCap data pipelines used to batch-upload patient records and convert outputs from diagnostic imaging systems into clinical datasets. Later expanded this work as Clinical Data Engineer.
- Built data cleaning and anonymization scripts to resolve naming conflicts in CSV datasets, remove unwanted variables, and ensure researcher access to filtered, privacy-compliant data.
- Led recruitment for 600+ patients at a 50% higher daily rate than the team average while maintaining 800+ high integrity patient files.

Laboratory Technician & QC Reviewer

CellCarta | September 2022 – March 2024

Blood Bank Medical Technologist

McGill University Health Centre (MUHC) | July 2022 – September 2022

Selected Projects

MCP Clinical Agent

Python, LangGraph, FastMCP, PostgreSQL, Ollama, Docker | 2026

- Built a local AI agent that turns natural language clinical questions into PostgreSQL queries, supports follow-up questions, recovers from SQL errors, and exports results to CSV. Achieved 20/20 on two end-to-end benchmark runs.

RAG Loremaster

Python, ChromaDB, Ollama/Gemini API, Docker | 2026

- Built a retrieval augmented generation system that crawls wikis and answers questions with citations. Indexed 4,701 pages into 86,784 vectors and scored 17/21 (81%) on a manual benchmark.

Clinical Medication Parser

Python, Ollama, Pydantic, pandas | 2026

- Built a two-stage local LLM pipeline that extracts medications from unstructured clinical notes into validated structured data with source text evidence. Achieved approximately 87% case-level quality across 30 cases.

Education

McGill University – Montréal

B.Sc., Computer Science (Artificial Intelligence) | Expected May 2028

- Minor: Entrepreneurship for Science Students

Dawson College – Montréal

Diploma of College Studies (DEC), Biomedical Laboratory Technology | May 2022